

Assignment 3: Used Car Price Prediction (summary)

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dataset	Vehicle dataset from CarDekho (kaggle), CAR DETAILS FROM CAR DEKHO.csv, 4,340 listings
my subset	roll number 150096724029 mod 3 = 0, so Petrol cars: 2,123 listings
cleaning	406 duplicate listings dropped (1,717 left), car_age = 2020 - year, brand and model from name, categories one-hot encoded (fuel is constant, so it drops out)
final model	Linear Regression: log(price) from car age, km driven, transmission, owner, seller type, brand and model, on 1,667 cars after removing the extreme 1%

results (80/20 split)

step	test R ²	5-fold CV R ²
v1 handout features, price as is	0.418	0.350
v2 log of price	0.621	0.629
v3-v4 + outliers removed + seller type	0.664	0.643
v5 + brand	0.703	0.722
v6 + model (final)	0.840 (train 0.868)	0.848 ± 0.016

final test RMSE 0.89 lakh, MAE 0.55 lakh (median test price 2.5 lakh). **pass condition (test R² >= 0.75): passed**, train-test gap 0.028.

what reduced resale value the most

- **age:** -8.6% for every extra year (holding everything else fixed). on a log scale price falls in a straight line with age
- **gearbox:** a manual sells for 14.2% less than the same car as an automatic
- **seller:** individuals get 10.9% less than dealers
- **km driven:** -2.1% per 10,000 km, smaller than it looks, because high-km cars are mostly old cars too
- **which car it is:** brand and model took test R² from 0.66 to 0.84

why the model performs as it does

cars lose value in percentages, so a straight line on raw rupees fits badly (0.42). log(price) makes the age effect a straight line, and brand and model tell the model what kind of car it is. limits: models seen fewer than 10 times are grouped as “other”, and the typical error is about 55,000 rupees.